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Next-Generation EMR System: Blockchain and Multimedia Communication Technologies

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Abstract— Hospital-centric data management has long created a situation where patients remain dependent on individual institutions for access to their own clinical histories. Record portability between hospitals is limited; server outages or cyberattacks can result in irreversible data loss; and multimedia clinical files such as imaging scans and consultation recordings fall outside what conventional text-based EMR architectures can accommodate. This paper describes the design and evaluation of a decentralized Electronic Medical Record platform constructed on Ethereum-compatible smart contracts together with IPFS-based off-chain storage. The system separates content storage from access governance — actual medical files are kept on IPFS through Pinata, whereas only their content identifiers are anchored on-chain. Six Solidity contracts partition responsibility across doctor management, patient records, hospital registration, insurance processing, research consent, and system administration. Patients authenticate via MetaMask wallets and determine, at the granularity of individual transactions, which clinicians may retrieve which records. Access requests from insurance entities and research bodies are processed only upon explicit patient authorization. Prototype evaluation spanning seven file formats — MP4, DICOM, JPEG, PNG, WAV, M4A, and PDF — recorded an average retrieval rate of 0.93 MB/s, an upload rate of 0.72 MB/s across 314 KB of test data, and a cumulative efficiency ratio of 92.54. The results indicate that off-chain multimedia storage paired with on-chain access governance can satisfy both functional security requirements and practical throughput constraints in healthcare settings.

Keywords – IPFS, EMR, Decentralization multimedia, Blockchain

I. INTRODUCTION

Healthcare Digital health infrastructure has grown rapidly across hospitals and clinics, yet the underlying storage model for patient records has barely changed data still lives on servers owned by individual institutions, cut off from other hospitals and fully opaque to the patient. Electronic Medical Records (EMR) act as structured digital repositories of a patient's clinical journey, holding diagnoses, laboratory results, prescribed medications, and medical imaging. The problem is not the concept of an EMR but its implementation: nearly every deployment is centralized within a single hospital's control as shown in fig 1, which creates

fragmentation of records across institutions, poor interoperability, and exposure to localized failures or attacks [1]. Multimedia data adds a further dimension to this challenge. MRI sequences, surgical recordings, and diagnostic audio cannot be efficiently stored or retrieved through text-oriented database systems, yet their clinical importance continues to rise.

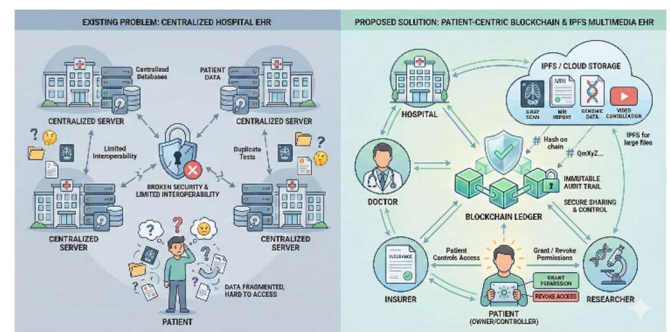


Figure. 1. Traditional vs Proposed Blockchain-Based EHR System

Smart contracts offer a technically concrete path through these problems. Deployed on a blockchain network, they are programs whose execution logic is fixed at deployment and no administrator can silently modify access rules after the fact. Every transaction is cryptographically signed and forms part of a permanent, auditable chain. Multiple hospitals can read from and write to the same distributed ledger under defined permission constraints, which provides a shared trust infrastructure that centralized databases cannot replicate. Blockchain's relevance in this domain stems precisely from the gap that conventional databases leave: patient-controlled, cross-institutional, tamper-evident access to medical records.

This paper proposes a blockchain-based multimedia EMR architecture that combines Ethereum smart contract access governance with decentralized file storage through the InterPlanetary File System. Five stakeholder categories such as patients, doctors, hospitals, insurance providers, and research institutions all interact through role-specific interfaces. Large multimedia files are stored on IPFS while their cryptographic hash references are recorded on-chain, preserving data integrity without inflating blockchain storage costs.

A. Organization of Paper

The Section II surveys related work. Section III presents the proposed system architecture and implementation detail. Section IV reports experimental results and performance

analysis. Section V concludes with directions for future development.

II. RELATED WORK

Recent advancements in blockchain technology and multimedia communication have significantly influenced secure healthcare data management systems. Several researchers have proposed different models focusing on data privacy, decentralized storage, and cross-institutional medical data sharing. However, many challenges such as patient-controlled consent, multimedia scalability, and multi-stakeholder workflows still remain open.

Mahajan and Junnarkar [2] integrated lightweight elliptic curve cryptography with a private blockchain to protect biomedical images across edge, fog, and cloud computing tiers. Encryption efficiency is demonstrably strong. The critical gap is that patient authorization is entirely absent: permission management resides with institutions rather than with patients, which reproduces the centralized trust model in a distributed guise. No mechanism exists for patients to grant or withdraw access independently.

Wu, Dwivedi, and Srivastava [3] proposed a theoretical framework coupling the POPPAC access control model with blockchain to protect private medical information. The conceptual design is well-argued. Its limitation is practical: no prototype was built, insurance and research organization workflows are unaddressed, and cloud storage is assumed throughout — a choice that sacrifices decentralization at the storage layer even while blockchain handles the access layer.

Zhao et al. [4] introduced StuChain, a student e-portfolio platform using a hybrid RBAC-ABAC access model, with demonstrated scalability. The architecture cannot transfer cleanly to healthcare. Patient consent management, clinical multimedia storage, and insurance claim processing are structurally absent from StuChain's design, not incidentally omitted. Adapting it would require near-complete redesign of its core permission model.

Mole and Shaji [5] deployed an Ethereum and IPFS-based EHR system and reported favorable throughput and latency figures. A significant architectural concern, however, is that patient records are maintained on a local Ganache development server — a tool designed for testing, not production. This choice reintroduces centralized data custody through the backend, while multimedia clinical files and research workflows fall entirely outside the system's scope.

Shuaib et al. [6] built a permissioned blockchain EHR sharing system using IBFT consensus and threshold signatures, achieving measurable improvements over proof-of-work alternatives. Despite this, permission control rests with miners and care providers rather than patients. A patient cannot independently revoke access; institutional mediation remains necessary, and insurance-related workflows are entirely absent from the implementation.

Landolsi, Hlaoua, and Ben Romdhane [7] produced a thorough survey of NLP-based information extraction from clinical documents, covering rule-based, statistical, and transformer-based approaches including BioBERT. The extracted information itself carries no protection in this

survey context — there is no access control mechanism and no tamper-proofing, so the extracted output could be retrieved or altered by unauthorized parties.

Chelladurai, Pandian, and Ramasamy [8] designed a patient-centric EHR system using blockchain-stored metadata hashes and tamper-evident access logs. Integrity verification is addressed well. The system does not extend to research consent management, insurance processing, or decentralized multimedia storage — coverage that any production-ready clinical platform would require.

Puneeth and Parthasarathy [9] reviewed more than 150 blockchain-based EHR papers, identifying interoperability, scalability, and access control as persistent open challenges. The value of this contribution is its systematic mapping of the field. Its limitation is that it proposes no engineering solution to any of the problems it documents.

Looking across these works, a clear gap emerges existing systems Across these works, a recurring structural gap is visible. Systems that prioritize security often do so at the cost of patient agency. Systems that emphasize decentralized storage rarely address the full set of stakeholders involved in clinical data flows. Patient-controlled consent, verified multimedia storage, and coordinated support for hospitals, insurers, and researchers have not been integrated into a single deployable architecture. The present work addresses that gap directly, through six coordinated smart contracts operating over IPFS-based distributed storage.

III. PROPOSED SYSTEM

A. Existing System

- **Centralized Storage and Limited Security:** Present EMR deployments rely on institution-managed databases — an architecture that concentrates risk at single points. Unauthorized access, data tampering, and server-level failures are not hypothetical edge cases; they represent documented recurring incidents across healthcare systems globally. Cross-institutional record verification is additionally impeded by the absence of any shared trust layer, which forces hospitals onto incompatible data schemas with no standard mechanism for interchange.
- **Multimedia Handling and Access Control Gaps:** Traditional healthcare systems struggle with securely storing multimedia medical files, lack patient-controlled access, and offer no structured role-based permission management for doctors, hospitals, insurers, and researchers — all of which the proposed system resolves through blockchain-based immutable storage, IPFS-based decentralized multimedia management, and smart contract-driven access control.

B. Scope of the System

The proposed platform targets four functional objectives simultaneously. First, it enables cross-institutional data sharing without requiring direct trust between hospitals, by anchoring access governance in a distributed

ledger rather than in any single institution's server. Second, it places access permission management directly in patients' hands at the transaction level. Third, it provides secure, content-addressed storage for large multimedia medical files through IPFS, keeping on-chain storage costs bounded. Fourth, it automates role verification, record creation, and consent processing through smart contracts, replacing manual administrative workflows with deterministic code execution.

C. Materials and Methods

The architecture integrates three coordinated layers: a React-based frontend, an Ethereum smart contract network, and IPFS-based distributed file storage, with MetaMask handling transaction signing throughout. The structural decision to decouple file storage from access metadata was foundational.

Storing binary medical files on-chain is both technically impractical, file sizes exceed reasonable block data limits and economically unsustainable given gas cost structures. IPFS resolves this by assigning each uploaded file a content identifier (CID) derived from a cryptographic hash of the file's contents. That CID, rather than the file itself, is written to the blockchain, providing an unforgeable reference that simultaneously functions as an integrity checksum.

The system block diagram as shown in fig 2 spans five layers. The User Layer encompasses all role-based stakeholder interaction. The Application Layer runs the React single-page interface with separate dashboard views for each role. The Blockchain Layer hosts the six Solidity contracts on Ethereum. The Multimedia Storage Layer is managed through IPFS via Pinata's pinning service. The Verification Layer operates at retrieval time, re-hashing downloaded files and comparing results against the stored CIDs to detect any modification.

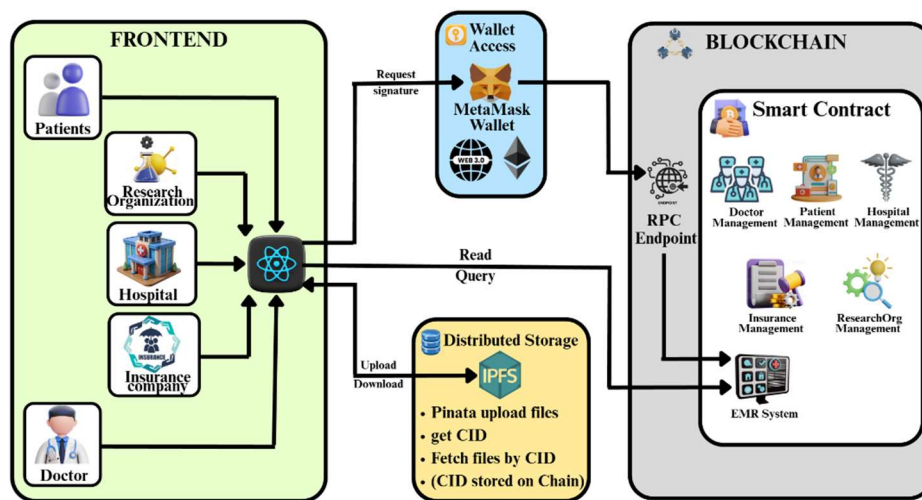


Figure 2. System Architecture of the Multimedia EHR System

Frontend Layer: The React interface provides distinct role-specific dashboards rather than a single filtered view. This separation is not cosmetic, conflating views at the interface layer risks blurring the permission boundaries enforced at the contract layer. Doctors access record creation interfaces; patients view and control their permission logs; administrators manage registration and approval queues. The frontend listens to smart contract events, updating dashboard state in real time without manual refresh.

Wallet and Transaction Layer: Authentication is handled entirely through MetaMask, eliminating any server-side session state or password database. Each stakeholder action generates a blockchain transaction signed by the user's private key before submission. This makes every recorded action cryptographically attributable and removes the login server as a compromise surface. The absence of a centralized credential store is itself a security property.

Blockchain and Smart Contract Layer: The Ethereum network provides transaction immutability and deterministic contract execution. Smart contracts manage role registration, record metadata, and access permissions. Transactions, once confirmed, cannot be edited and a property that makes the audit trail tamper-proof by construction rather than by policy. Contract-emitted events propagate to the frontend, enabling live dashboard updates without polling.

Distributed Storage Layer: Multimedia files are uploaded to IPFS through Pinata. Pinata's pinning service ensures that content remains persistently available and without active pinning, IPFS nodes have no obligation to retain files that no node is actively serving. Upon upload, IPFS returns the file's CID, which is then stored on-chain. Retrieval reverses this: the authorized user requests the CID from the contract, fetches the file from IPFS, and the system re-hashes the result. Any file modification produces a different hash, failing comparison against the on-chain record and flagging the retrieval as compromised.

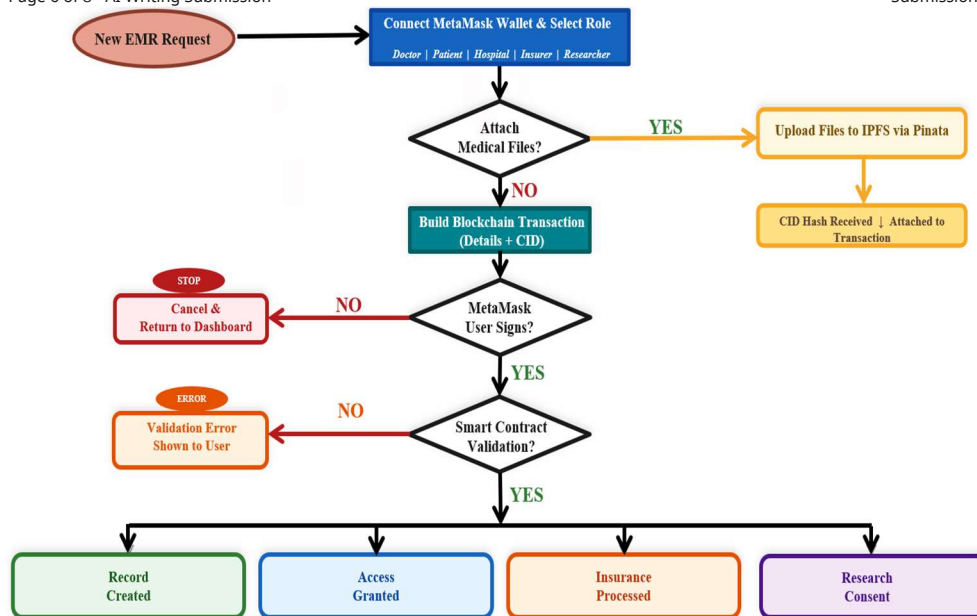


Figure 3. Flow Chart of the Multimedia EHR System

Workflow Description

Users log in through the React-based interface using their MetaMask wallet, where each stakeholder sees a dashboard based on their role. Doctors can create records, patients can manage access permissions, and administrators can approve registrations. When multimedia files such as scans, videos, or audio are uploaded, they are stored in IPFS through Pinata, which generates a unique CID for each file. This CID, along with the record details, is then sent as a transaction to the smart contract. Before submission, MetaMask asks the user to review and sign the transaction. The smart contract verifies the user's role and permissions, then securely stores the record metadata and CID on the blockchain. After confirmation, the dashboard updates automatically, and authorized users can retrieve files from IPFS using the stored CID, while integrity is ensured through hash verification. This process maintains a secure and tamper-proof audit trail for all medical records.

Algorithm 1:

Stepwise SHA-256 Hash Generation for Medical File

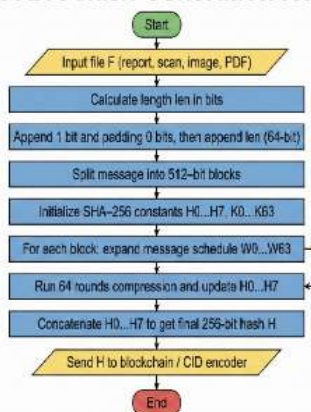


Figure 4. SHA Algorithm

Each file entering the system is passed through SHA-256 before IPFS storage, producing the deterministic hash that

becomes the CID input. The algorithm processes files as follows: the input bit stream is padded to a length congruent to 448 bits modulo 512, with a trailing 1 bit, zero bits, and the original 64-bit message length appended. This padded message is divided into 512-bit blocks. Eight 32-bit state variables (H0 through H7) are initialized to constants derived from the first eight prime numbers. Each block undergoes sixty-four rounds of compression using bitwise rotation, XOR, and mixing operations that update the state variables continuously. Final concatenation of H0 through H7 produces the 256-bit hash output, which the IPFS CID encoder transforms into the content identifier anchored on-chain [10]. Any modification to the original file produces a detectably different hash.

D. Dataset used

The evaluation dataset was assembled from multiple open healthcare repositories to represent the practical range of multimedia record types encountered in clinical settings. DICOM, JPEG, and PNG imaging files were obtained from Zenodo's medical imaging collections and standard radiology repositories. PDF clinical documents, prescriptions, diagnostic reports, and discharge summaries were sourced from the PhysioNet MIMIC-III clinical database, which is widely used in biomedical informatics research. MP4 video files documenting surgical and diagnostic procedures were drawn from medical video datasets hosted on Hugging Face. WAV audio recordings simulating patient consultation exchanges came from publicly accessible healthcare communication datasets. The assembled set represents the categories a real hospital EMR must handle radiology, documentation, procedural video, and patient interaction audio sourced from repositories available to academic researchers.

E. Medical record transmission and verification under multimedia communication technology.

In the proposed system, when a clinician uploads a medical file, it is stored on IPFS through Pinata, which generates a unique CID for that exact file. This CID is then recorded on the blockchain along with details such as the uploader and timestamp. When an authorized user wants to access the file, the system retrieves it from IPFS using the stored CID and verifies its integrity through hash matching. If the file has been changed in any way, the verification fails. This ensures that medical records remain original, secure, and tamper-proof during transfer without relying on any central authority[11].

F. Design and implementation of smart contracts

The proposed system uses six separate Solidity smart contracts on Ethereum, with each contract handling a specific function. This modular design improves security and makes the system easier to manage and update. The EMRSystem contract acts as the central coordinator by connecting the other contracts and managing admin roles. DoctorManagement handles doctor registration and allows approved doctors to create or update patient records. PatientManagement stores patient details and lets patients control who can access their records. HospitalManagement verifies hospitals and links doctors to their institutions. InsuranceManagement manages insurance claim requests, while ResearchOrganizationManagement handles patient consent for research data access. Core operations such as record creation, access requests, insurance requests, and approval workflows are carried out through these contracts, ensuring that the frontend stays updated with the latest system activity.

IV. RESULTS AND DISCUSSION

A. System and Software Requirements

The development and testing environment required a machine with a multi-core processor at 2.0 GHz or above, 8–16 GB RAM, and at least 20 GB of available storage to run the full stack concurrently web application, local Ethereum node, and IPFS services. Frontend development used React with Node.js as the runtime. Smart contracts were written in Solidity and tested locally against a Ganache node through the Hardhat development framework. IPFS integration was managed through the Pinata API. Transition from local Ganache testing to an Ethereum testnet validated the system's behaviour under realistic network conditions.

B. Experiment and performance analysis

a) Design of experimental dataset and multimedia medical record

Table 1. Experimental Multimedia record dataset design

Data type	Medical records	Data size (GB)	Data content
Plain text	Medical history	1.34	Patient details, prescriptions
Image	Diagnostic scan images	2.2	X-ray, MRI, CT scan images
Video	Procedure	5.6	Surgery or

	recording		diagnostic procedure videos
Voice Recordings	Consultation audio	1.23	Doctor consultation recordings
PDF	Medical reports	0.76	Laboratory reports, discharge summary

The dataset was chosen to reflect the practical diversity of clinical multimedia rather than benchmark-optimized synthetic data. These records represent the categories most frequently encountered in hospital EMR systems. Evaluating across this range rather than focusing on a single format that provides a meaningful basis for assessing the system's general-purpose viability.

b) Evaluation indicators

Table 2. Result of the system performance evaluation

Metrics	Plain text	Images	Video	Voice Recordings
Throughput(sec)	1.2	1.8	2.6	2.1
Response time (ms)	420	560	740	610
Execution Time of contracts (s)	0.35	0.42	0.48	0.44
On-chain Overhead (%)	3.4	6.3	9.7	7.1

We evaluated throughput, response time, smart contract execution time, and on-chain overhead to see how the system performs with different types of data. Larger files like videos (2.6 sec) and voice recordings (2.1 sec) take longer than plain text (1.2 sec) because they involve more data. However, smart contract execution stays almost constant (0.35–0.48 sec), showing it's not the main bottleneck. Instead, delays mainly come from IPFS upload time and Ethereum block confirmation. On-chain overhead also increases with file size, reaching about 9.7% for videos while staying under 3.5% for text.

c) Performance comparison between based on File type

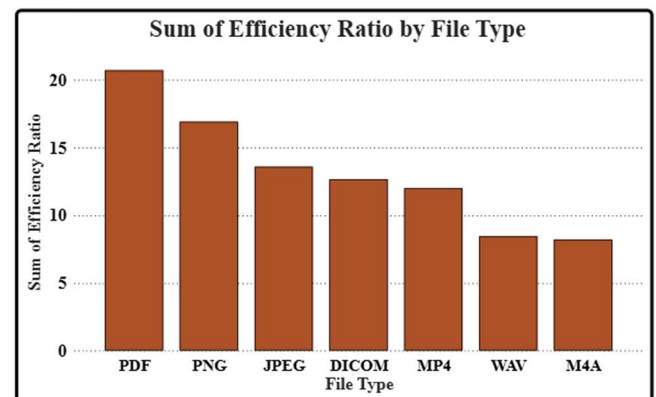


Figure 5. Efficiency ratio

The graph shows the efficiency ratio of different multimedia file types used in the system. Document and image formats achieve higher efficiency, while audio formats show comparatively lower efficiency due to larger processing requirements.

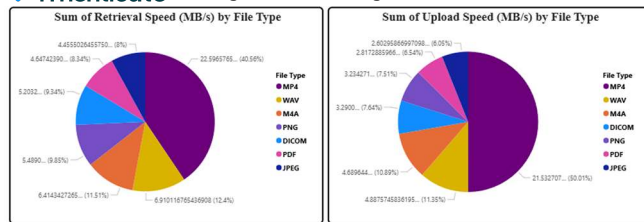


Figure 6. Statistical Breakdown

The chart shows retrieval speed for different multimedia file types, where **MP4** files have the highest share while other formats show moderate performance. The other shows upload speed distribution across file types, with **MP4** contributing the highest portion compared to other formats.

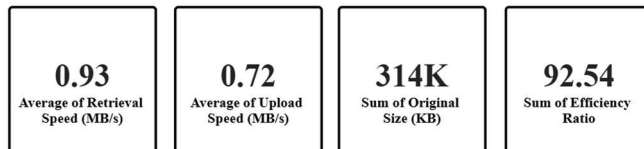


Figure 7. KPI cards summarizing system efficiency.

This indicate an average retrieval speed of 0.93 MB/s and an average upload speed of 0.72 MB/s for multimedia medical records in the system. The dataset contains a total original size of 314 KB, with an overall efficiency ratio of 92.54, demonstrating efficient storage and retrieval performance.

V. CONCLUSION AND FUTURE SCOPE

This work addressed a key gap in existing EMR systems by developing and testing a decentralized platform that combines patient-controlled access with multimedia medical record storage. The system connects patients, doctors, hospitals, insurers, and research organizations through role-based interfaces, while permission control is enforced through smart contracts instead of a centralized authority. Record creation, access approval, and transaction validation are performed on-chain, making the stored data tamper-resistant. Clinical files are stored on IPFS, while only their content identifiers are recorded on the blockchain to reduce storage cost and preserve integrity. Prototype evaluation across seven multimedia file formats confirmed the practical usability of the system. The platform achieved an average retrieval rate of 0.93 MB/s, an overall efficiency ratio of 92.54, and contract execution times of less than 0.5 seconds across all record types. Even for video records, the on-chain overhead remained below 10%, validating the effectiveness of the off-chain storage design. Future work will focus on integration with hospital systems, HL7 FHIR interoperability, stronger file-level encryption, mobile accessibility, and improved scalability. The architecture can also be extended to support automated insurance verification and secure, consent-based research data sharing.

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